
GreenLoop F2C — Executive Summary

1. The Opportunity

Singapore imports over 90% of its food. The COVID-19 pandemic and 2022 egg shortage caused price spikes lasting months. The government commits SGD 70M through the ACTF fund to shorten supply chains. Swiss vertical-farm competitor Greenphyto AG raised CHF 12M in January 2025 citing Singapore as their APAC entry point. GreenLoop is first to build the AI-native operating system for hydroponics — competitors plan farms by hand.

2. The Product

GreenLoop F2C is a 5-layer AI platform managing a vertical-hydroponic farm from seed to delivery. Layer 1 (XGBoost quantile regression) predicts demand with confidence intervals using 9 indoor-farm-appropriate features: lag shipments (7/14/28d), rolling 28d mean/std, cyclical week encoding, day-of-week, Singapore holiday flags, consumer rainy-day signal (simulated by monsoon month), indoor climate readings (temperature, humidity, CO2 deviation from 800 ppm optimal), electricity tariff tiers, and cross-crop market density. Layer 2 (OR-Tools MILP) generates the profit-maximising daily operating plan in **< 50 ms** — enabling real-time re-planning when conditions change. Layer 3 (PPO RL via Gymnasium) autonomously controls climate. Layer 4 (K-Means k=4 + UMAP) segments customers. Layer 5 (RAG with ChromaDB + Claude) answers investor questions live.

Key differentiators:

- **30/30 customers served** by OR-Tools CVRPTW in 6-hour compressed window (nearest-neighbor fails 20-40% of time windows)
- **Silhouette 0.765** at k=4 — beats k=3 (0.698) and k=5 (0.742) for customer segmentation
- **Typhoon scenario (v2)**: live dashboard button cuts delivery window 12 h to 6 h; 30% probability of power outage triggers 4-hour UPS countdown with emergency harvest protocol; demand surge +20% from panic buying; MILP re-solves and shows profit delta in milliseconds



Figure 1: Computer vision diagnosis from EfficientNet-B0 dual-head transfer learning — detecting growth stage and nutrition status from rack photography.

3. How It Works

6 ML Techniques Across 4 Course Domains

#	Technique	MGMT 655 Week
1	XGBoost quantile regression (q=0.05/0.50/0.95)	Week 2-3
2	OR-Tools MILP (CP-SAT) — 10 tiers, 24h LED, tariff-aware	Week 4
3	K-Means k=4 clustering + UMAP visualization	Week 5
4	EfficientNet-B0 dual-head CNN (growth + nutrition)	Week 6-7
5	PPO reinforcement learning — Gymnasium simulation	Week 7
6	RAG (ChromaDB + sentence-transformers + Claude)	Week 5-6

Daily flow: 05:00 Layer 1 demand forecast (q=0.95 upper CI) to 06:00 Layer 2 MILP plan (< 50 ms) plus Layer 2b VRP routes (30 customers) to 06:00-22:00 Layer 3 PPO RL controls climate to 08:00-18:00 deliveries to daily Layer 4 re-segmentation to on-demand Layer 5 RAG.

Typhoon proof: The Typhoon button triggers a full resilience cascade — not a delivery delay simulator. A 6-hour delivery window compression triggers 30% probability of power outage. If the grid fails, a 4-hour UPS countdown begins with a live red-banner timer in the dashboard. The PPO RL agent forces LEDs off and takes a –500 reward penalty per step on battery. Layer 2 MILP simultaneously re-solves in emergency harvest mode: 12 crops harvested early to prevent total loss if power does not return, cold storage switches

activated for unsold produce. Meanwhile, rainy weather keeps consumers indoors — Layer 1 applies a +20% demand surge to tomorrow’s plan. Every layer responds autonomously. MILP re-solves in under 50 ms — proving the farm re-plans itself under multi-variable stress without human intervention.

4. Governance & Production Readiness

Our system is not just a working demo — it is evaluated against production-readiness criteria:

Technical validation (Phase 7 Red-Team) - 38 adversarial tests across all 6 ML layers - Covers data poisoning, prompt injection, constraint stress, reward hacking, system-wide cascade - 100% passing; honest limitations documented in `journal/phase7-redteam.md`

Ship decision (Phase 8 Deployment Gate) - 5 gates, 25 criteria, automated evaluation via `run_deployment_gate.py` - Technical / Business / Risk / Compliance / Monitoring dimensions - Current judgment: **CONDITIONAL SHIP** for Phase 1 Pilot - Gate 2 Business Viability **CONDITIONAL** — pilot data required

Ethical analysis (Phase 5 Implications Audit) - 6 layers × 3 categories: data bias, decision bias, stakeholder impact - 1 HIGH severity identified (PlantVillage dataset bias → Asian crop misclassification) - 6 stakeholders mapped: net positive for 5, neutral for 1 (workers) - Integrated with Gate 4 Compliance — unmitigated HIGH/CRITICAL blocks deployment

Operational monitoring (Phase 13 Drift Monitoring) - 14 checks across 6 layers + system-wide - Feature drift: KS test on XGBoost inputs (weekly) - Performance drift: MILP infeasibility rate (hourly), EfficientNet confidence (daily), PPO reward ratio (weekly) - Concept drift: XGBoost prediction bias, PPO action distribution, segment stability, customer segment size, MILP constraint violations - YAML-scheduled execution from hourly to weekly

Together, 13 of 14 MGMT 655 phases completed. Retrospective documentation (analyze + redteam per phase in `journal/`) ensures transparency even where the COC workflow was compressed.

5. Business Model

Stream	Model	Year 1 ARR
B2B SaaS — Farm OS license	SGD 8K/rack/year	SGD 480 K
B2C subscription boxes	15% GMV commission	SGD 180 K
AI model licensing	One-time + support	SGD 60 K

Stream	Model	Year 1 ARR
Total		SGD 720 K

Unit economics: ACV SGD 8,000/rack/year; 20-rack farm; 82% gross margin; CAC/LTV ratio 0.15; payback 11 months. **GTM:** Phase 1 — 1 anchor farm, Jurong Innovation District, 3 paying accounts. Phase 2 — 10 farms (SG + Malaysia), B2C subscription, SGD 500 K ARR. Phase 3 — APAC AI model licensing, SFA national monitoring partnership.

6. Evidence & Validation

Live Dashboard Metrics (typical run)

Metric	Value
Revenue	~SGD 1,000-1,200 / day
Profit	~SGD 350-450 / day
MILP solve time	< 50 ms
Optimality gap	0% (proven optimal)

VRP — 30 Singapore customers, Jurong depot

Metric	Value
Customers served	30 / 30
Total route	147.1 km
Logistics cost	SGD 188.87

Sustainability (live-computed by Layer 2 plan): **95% less water** vs. conventional farming (2 L/kg vs. 20 L/kg; source: AVA Singapore 2019). **87% less CO2** (0.3 vs. 2.5 kg-CO2/kg; source: SFA 2023 lifecycle analysis).

Test suite: 434 tests — all governance, monitoring, adversarial, and unit tests passing. See [tests/](#) for full breakdown.

7. Team

Member	Focus
Takahide Kawabe	Farm OS, MILP, VRP, RL, full-stack integration
QN	Business model, segmentation, sustainability

Member	Focus
Claude (AI agent)	RAG pipeline, CC/CO knowledge extraction, testing

4-page dashboard fully functional: Farm OS (Layer 1-2 + Sustainability + Typhoon) | Logistics (CVRPTW map, 30/30 routes) | Retail AI (K-Means + UMAP, silhouette 0.765) | Media AI (RAG chatbot + demo mode).

8. The Ask

Phase 1 Pilot — SGD 80,000 for 20% equity

Amount SGD 80,000

Equity 20%

Duration 12 weeks

Farms 3 pilot farms (Jurong, Lim Chu Kang, Woodlands)

Use of funds: Hardware sensors (SGD 25K) · Singapore training data collection (SGD 20K) · Monitoring infrastructure (SGD 15K) · Pilot operations (SGD 12K) · Contingency (SGD 8K).

What happens next: Week 2 — install sensors on 3 pilot farms. Week 6 — first Singapore-specific model retraining with Phase 1 data. Week 12 — governance review and Phase 2 decision gate.

Submitted: 2026-04-24 | Greenloop-F2C | main | 434 tests passing | 13 of 14 phases complete